Chapter 14, Problem 77P

Problem

A series *RLC* circuit has *R* = 10 Ω, *ω*0 = 40 rad/s, and *B* = 5 rad/s. Find *L* and *C* when the circuit is scaled:

(a) in magnitude by a factor of 600,

(b) in frequency by a factor of 1,000,

(c) in magnitude by a factor of 400 and in frequency by a factor of 105.

Step-by-step solution

Step 1 of 4

L and C are needed before scaling

$$B=\frac{R}{L}\rightarrow L=\frac{R}{B}=\frac{10}{5}=2H_Z$$

$$\omega_o=\frac{1}{\sqrt{LC}}\rightarrow C=\frac{1}{\omega_o^2 L}=\frac{1}{(1000)(2)}=312.5\mu F$$

Comments (1)

Anonymous

w is given: 40 rad/sec so w^2 is 1600, not 1000

Step 2 of 4

(a) $L^{'}=K_m L=(600)(2)=1200\,H_Z$

$$C^{'}=\frac{C}{K_m}=\frac{3.125\times10^{-4}}{600}=0.5208\mu F$$

Step 3 of 4

(b) $L^{'}=\frac{L}{K_f}=\frac{2}{10^3}=2MH_Z$

$$C^{'}=\frac{C}{K_f}=\frac{3.125\times10^{-4}}{10^3}=312.5nF$$

Step 4 of 4

(c) $L^{'}=\frac{K_m}{K_f}L=\frac{(400)(2)}{10^5}=8MH_Z$

$$C^{'}=\frac{C}{K_m K_f}=\frac{3.125\times10^{-4}}{(400)(10^5)}=7.81PF$$